

Question 1 (2 points)

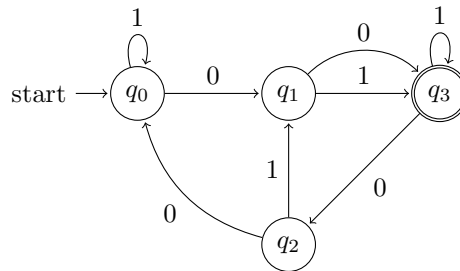
Let L be a regular language. Then according to the pumping lemma for regular languages, for some pumping length $p \geq 1$, every string $w \in L$ with $|w| \geq p$ can be written as $w = xyz$, such that:

- $|y| \geq 1$
- $|xy| \leq p$ and
 - A. For all $i \geq 0$, $xy^iz \in L$
 - B. For some $i \geq 0$, $xy^iz \in L$
 - C. For all $i \geq 1$, $xy^iz \in L$
 - D. x, y, z are uniquely determined by w

Option A is correct. Refer to the Pumping Lemma (Section 6.1 in Week 3 Lecture Notes)

Question 2 (2 points)

What is the number of states in the minimal DFA corresponding to the given DFA?



- A. 1
- B. 2
- C. 3
- D. 4

Option D is correct. Minimize the given DFA according to the process given in Section 7.1 in Week 3 Lecture Notes. You will find that none of the states belong to the same equivalence class at the end of the process. Therefore, the given DFA is already minimal.

Question 3 (2 points)

Which of the following is a **regular language**?

- A. $\{a^n b^n \mid n \geq 0\}$
- B. $\{a^m b^n \mid m, n \geq 0\}$
- C. $\{a^n b^n c^n \mid n \geq 0\}$
- D. $\{w \in \{a, b\}^* \mid w \text{ has equal number of } a\text{'s and } b\text{'s}\}$

Option B is correct. The regular expression corresponding to this language is a^*b^* . It can be shown using the pumping lemma that the other languages are not regular.

Question 4 (2 points)

Which of the following languages is **not regular**?

- A. $\{a^{2n} \mid n \geq 0\}$
- B. $\{a^n b^m \mid m \leq n\}$
- C. $\{a^{(n+1)^2 - n^2} \mid n \geq 0\}$
- D. $\{a^n \mid n \text{ is odd}\}$

Option B is correct. We can use pumping lemma to show that this language is not regular.

We will prove this via contradiction. Let's say this language (say, L) is regular. Then it must satisfy the pumping lemma for some integer $p \geq 1$. Let us consider the string $w = a^k b^k$ (such that $k > p$) which clearly belongs to this language. Now, however you decompose w as xyz (such that $|xy| \leq p$ and $|y| \geq 1$), y can consist of only a 's. If we consider the string xy^0z now, it will contain less number of a 's than b 's and hence will not belong to L . However, according to the pumping lemma, this string should belong to L if L is regular. This is a contradiction to our assumption that L is regular.

Question 5 (2 points)

Which of the following statements is **not true**?

- A. Every language that satisfies the pumping lemma is regular
- B. Every regular language satisfies the pumping lemma
- C. A language that does not satisfy the pumping lemma is not regular
- D. Some non-regular languages satisfy the pumping lemma

Option A is correct. Refer to the Pumping Lemma (Section 6.1 in Week 3 Lecture Notes)

Question 6 (2 points)

Which of the following languages is **not regular**?

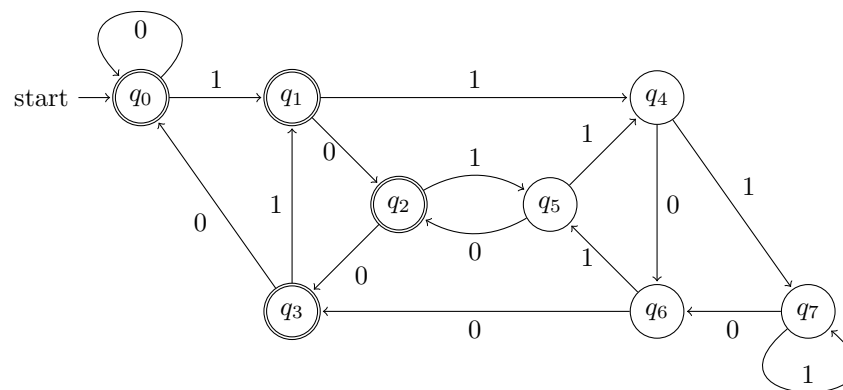
- A. $\{a^p \mid p \text{ is a prime number}\}$
- B. $\{w \in \{0,1\}^* \mid w \text{ contains at most two 1's}\}$
- C. $\{w \in \{0,1\}^* \mid w \text{ ends with } 00\}$
- D. $\{w \in \{0,1\}^* \mid \text{binary } w \text{ is divisible by } 5\}$

Option A is correct. We can use pumping lemma to show that this language is not regular.

We will prove this via contradiction. Let's say this language is regular. Then it must satisfy the pumping lemma. Therefore there exists an integer $p \geq 1$ such that for some $k \leq p$, 0^{n+kn} also belongs to the language, where n is a prime number. (The substring y in the pumping lemma has to be some string of 0's of length $k \leq p$ here and when we pump that substring y n times into the string 0^n , we get 0^{n+kn}). $n + kn = n(k + 1)$ is not a prime number. This is a contradiction.

Question 7 (2 points)

How many states are there in the **minimal DFA** equivalent to the following DFA?



- A. 3
- B. 4

C. 6

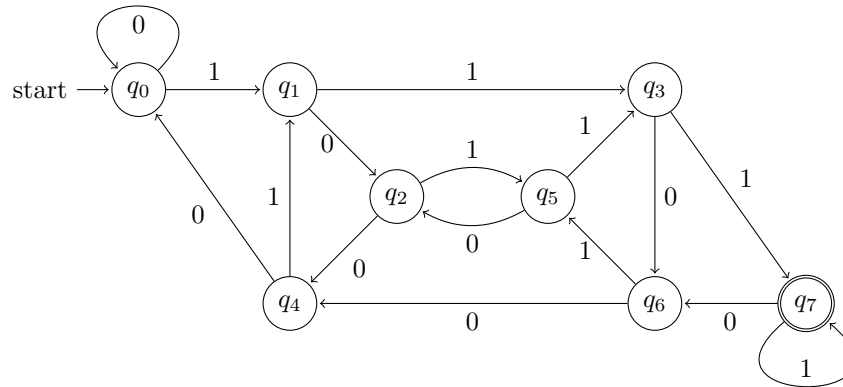
D. 8

Option C is correct.

Minimize the given DFA according to the process given in Section 7.1 in Week 3 Lecture Notes. You will find that there are six equivalence classes of states for this DFA $\{q_0, q_3\}$, $\{q_1\}$, $\{q_2\}$, $\{q_4, q_7\}$, $\{q_5\}$ and $\{q_6\}$. Therefore, we can construct a minimal DFA with 6 states corresponding to these 6 equivalence classes.

Question 8 (2 points)

How many **accepting states** are there in the **minimal DFA** equivalent to the following DFA?



A. 1

B. 2

C. 3

D. 4

Option A is correct.

Minimize the given DFA according to the process given in Section 7.1 in Week 3 Lecture Notes. You will find that there are four equivalence classes of states for this DFA $\{q_0, q_2, q_4, q_6\}$, $\{q_1, q_5\}$, $\{q_3\}$ and $\{q_7\}$. Out of these, only one correspond to the accept state- $\{q_7\}$. Therefore, in the minimal DFA also, there will be only one accept state.